Introducing smalltalk-dev-plugin

AI-Driven Development Toolkit for Pharo Smalltalk

Masashi Umezawa

https://github.com/mumez/smalltalk-dev-plugin

What is smalltalk-dev-plugin?

> A comprehensive toolkit that bridges **AI code editors** and the **Pharo Smalltalk** environment.

The AI Agent Problem for Smalltalk

Using AI coding agents with Smalltalk doesn't work well out of the box:

- **No knowledge of Pharo** — AI doesn't know how to interact with the environment
- **Unknown project structure** — Typical Tonel/package layout is unfamiliar
- **No testing/debugging skills** — Can't run tests or diagnose errors
- **No library access** — Can't browse existing classes or methods
- **Poor coding style** — Doesn't know Smalltalk idioms and conventions

> This is a major barrier for developers who want AI assistance in Smalltalk.

Enter smalltalk-dev-plugin

We gave AI the complete skillset for Smalltalk development:

Challenge	Solution
Environment interaction	MCP tools for Pharo communication
Project structure	`/st-setup-project` with templates
Testing & debugging	`/st-test`, `/st-eval`, debugger skill
Library navigation	Finder skills, code search tools
Code quality	Validator, linter, commenter agent

Result: AI becomes a reliable Smalltalk development partner!

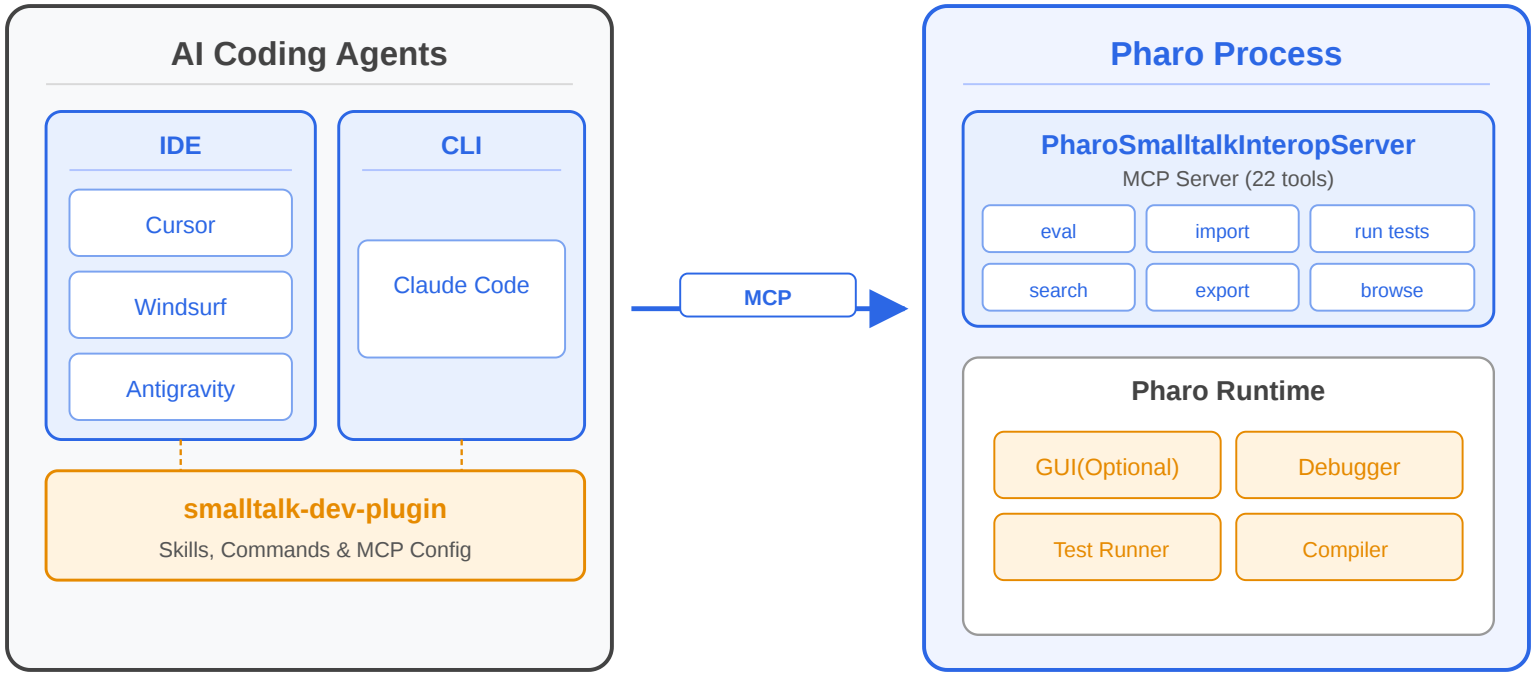
"AI editing as the source of truth"

> *Edit Tonel files in the AI editor, import to Pharo, test, and iterate.*

At a Glance

- 9 slash commands
- 5 specialized AI skills
- 22 MCP tools for Pharo interaction
- Automated code quality checks and documentation generation

Architecture



- Existing AI agents talk to Pharo via MCP — no custom IDE needed
- Even works with headless Pharo — usable in CI and remote agents like Devin

Supported AI Agents

Agent	Support Level
Claude Code	Full support (primary)
Cursor	Simplified setup via script
Windsurf	Simplified setup via script
Antigravity	Simplified setup via script
GitHub Copilot CLI	Simplified setup via script
OpenCode	Simplified setup via script

The rest of this presentation focuses on the Claude Code applied version.

Installation

Installation (1) — Install the Plugin


```
# Add marketplace from GitHub
claude plugin marketplace add mumez/smalltalk-dev-plugin

# Install the plugin
claude plugin install smalltalk-dev
```

Installation (2) — Set Up Pharo Side

Option A: Docker (Easy)

Use smalltalk-interop-docker to run a pre-configured Pharo image:

```
docker compose up -d
```

Option B: Local Pharo

Install and start the interop server manually:

```
Metacello new
  baseline: 'PharoSmalltalkInteropServer';
  repository: 'github://mumez/PharoSmalltalkInteropServer:main/src';
  load.

SisServer current start.
```

Installation — Other AI Agents

Setup scripts are provided for each agent:

```
./extra/setup-cursor.sh [target-directory]
./extra/setup-windsurf.sh [target-directory]
./extra/setup-antigravity.sh [target-directory]
./extra/setup-copilot.sh [target-directory]   # GitHub Copilot CLI
./extra/setup-opencode.sh [target-directory]  # OpenCode
```

Limitations

- Some features may differ depending on the agent
- Pharo-side setup is the same as for Claude Code

Basic Usage

Where to Launch Claude

Run `claude` in an **empty directory** or your **existing project directory**.

- You can also point to an Iceberg repository directory
- However, to avoid conflicts, it is better to **clone the repo into a separate directory** and use that instead

/st-buddy — Main Entry Point

`/st-buddy` is a friendly assistant that routes to the right tool.

Just describe what you want in natural language:

```
/st-buddy I want to create a Person class with name and age
```

The assistant will:

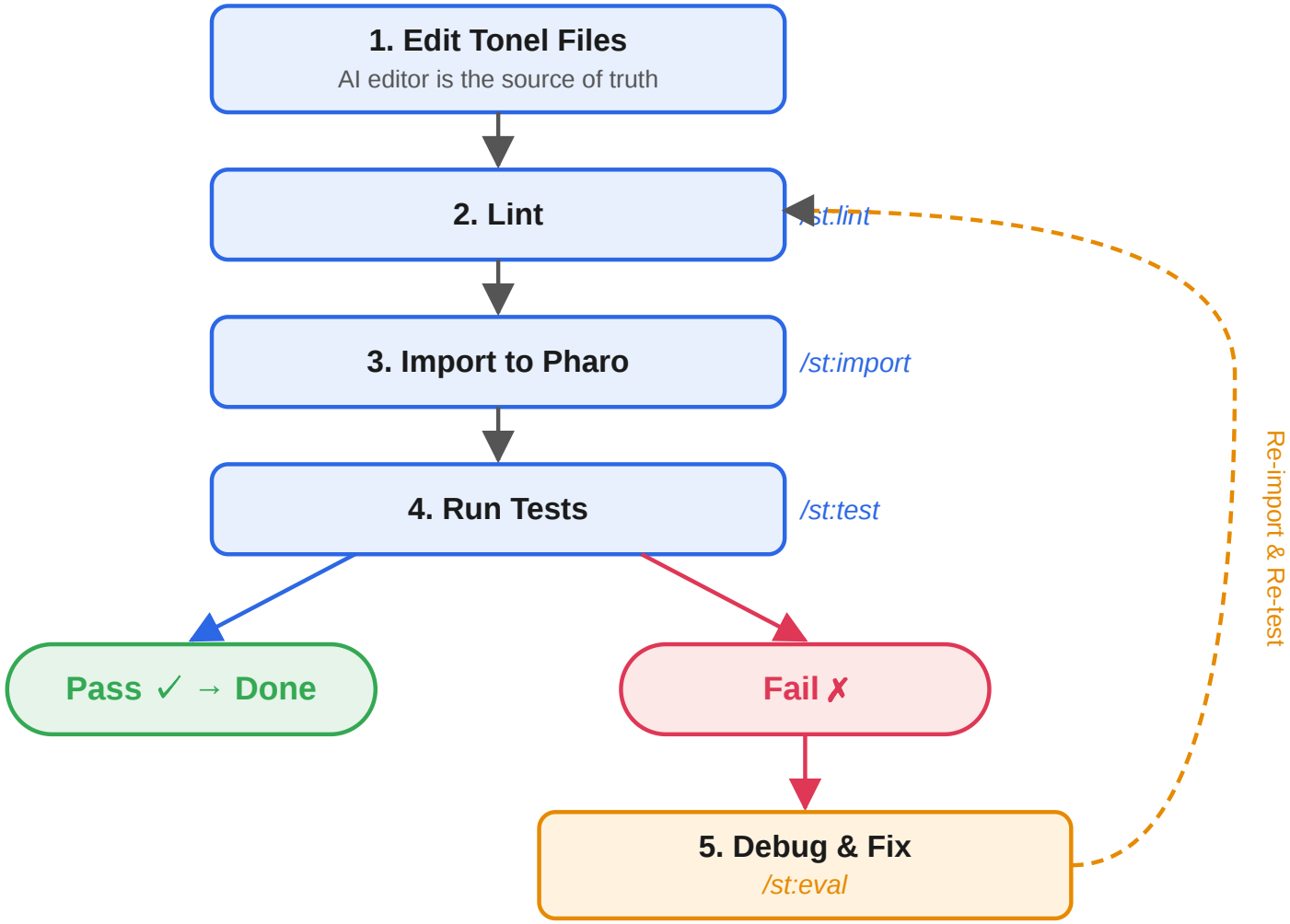
1. Load the appropriate skill automatically
2. Create the Tonel files
3. Guide you through linting, importing, and testing

/st-buddy — Intelligent Routing

Your Intent	Loaded Skill
"Create...", "Add..."	smalltalk-developer
"Error...", "Test failed..."	smalltalk-debugger
"How do I use...?"	smalltalk-usage-finder
"Who implements...?"	smalltalk-implementation-finder

You don't need to remember which tool to use — just talk to `/st-buddy`.

Development Workflow



Components

Skills (1) — Developer & Debugger

smalltalk-developer

The core development skill. Knows Tonel format, package structure, and the full Edit → Lint → Import → Test workflow.

smalltalk-debugger

Systematic debugging specialist. Diagnoses errors, runs incremental code execution with `/st-eval`, and guides you to the fix.

Skills (2) — Code Finder

smalltalk-usage-finder

Discovers how classes and methods are used. Finds examples, analyzes usage patterns, and helps you understand unfamiliar APIs.

smalltalk-implementation-finder

Finds method implementations across class hierarchies. Useful for learning coding idioms and assessing refactoring impact.

Skills (3) — Documentation

smalltalk-commenter

Automatically suggests **CRC-style class comments** for undocumented classes. Triggered after file edits via hooks — non-intrusive and prioritizes classes that need documentation most.

Slash Commands

Command	Purpose
`/st-buddy`	Friendly assistant (main entry)
`/st-init`	Initialize development session
`/st-setup-project`	Create project boilerplate
`/st-eval`	Execute Smalltalk expressions
`/st-import`	Import Tonel packages to Pharo
`/st-export`	Export packages from Pharo
`/st-test`	Run SUnit tests
`/st-lint`	Check code quality
`/st-validate`	Validate Tonel syntax

MCP Servers

Two MCP servers power the integration behind the scenes:

pharo-interop (22 tools)

Pharo communication — eval, import/export, tests, code navigation, etc.

smalltalk-validator (5 tools)

Tonel validation and linting.

Note: You rarely need to interact with these directly.

`/st-buddy` handles orchestration for you.

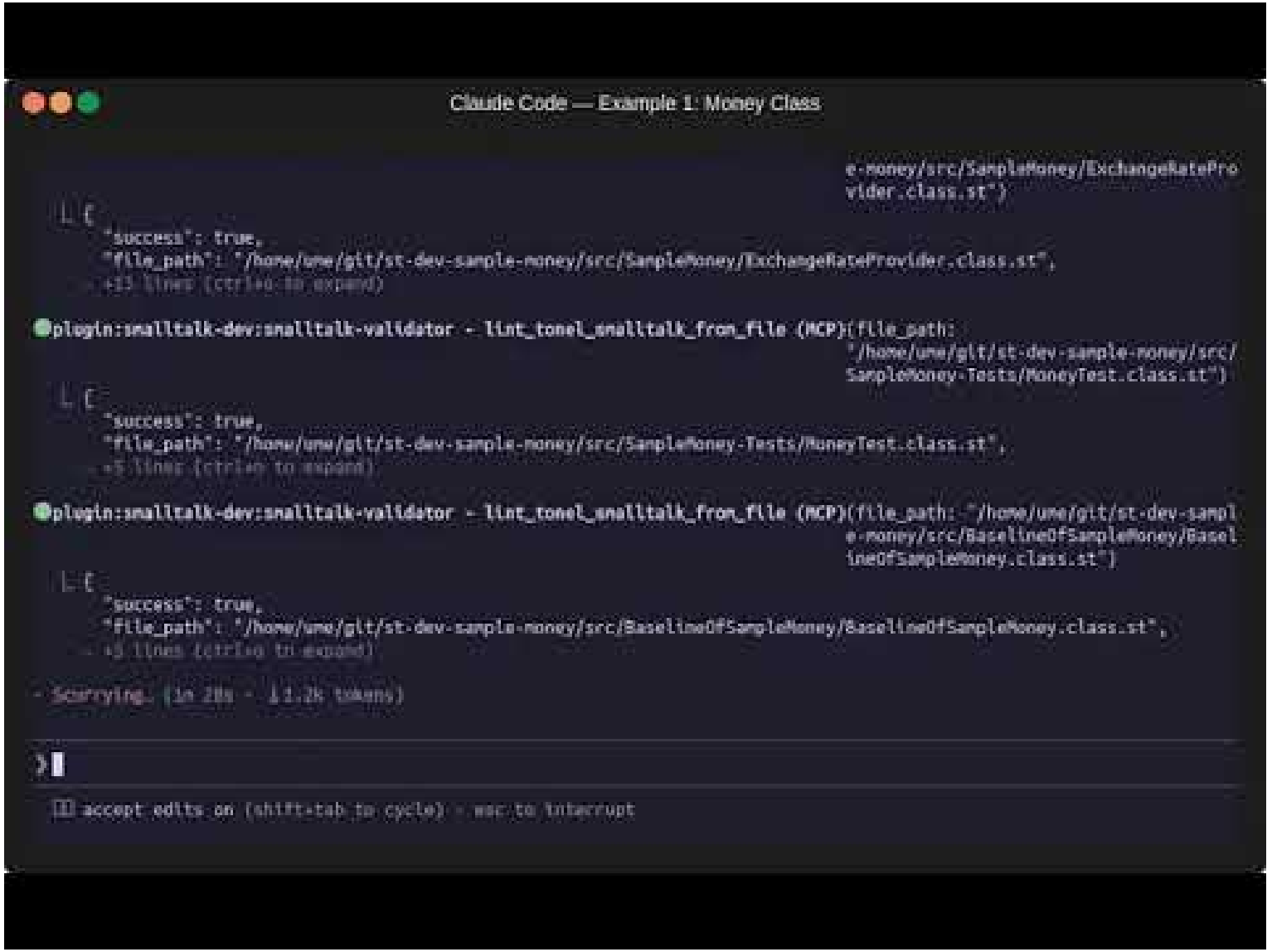
Live Examples

Example 1 — Starting from Scratch

A simple use case: starting from an empty directory as a Smalltalk beginner.

```
/st-buddy I'm a Smalltalk beginner. I want to create a Money class
that can perform arithmetic operations across different currency units.
Please help me from the project initialization.
```

Example 1 — What Happens Next



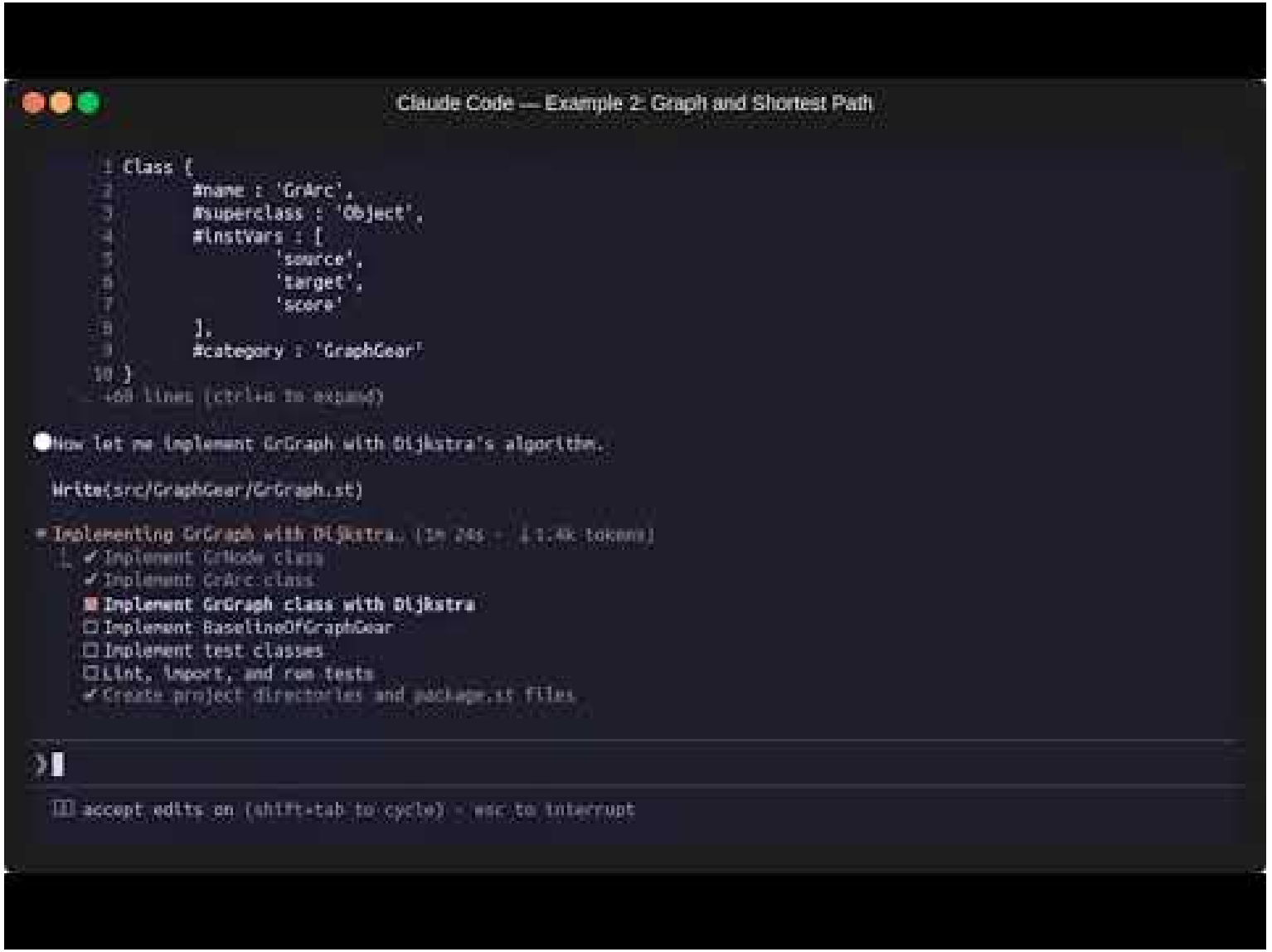
- Video
- Generated source
- Demo

Example 2 — Graph Algorithms

A more complex use case for someone familiar with Smalltalk.

```
/st-buddy I want to create GrNode and GrArc to represent directed graphs
and solve shortest path problems. Nodes have a name, arcs have a score.
Let's start this as a GraphGear project.
```

Example 2 — What Happens Next



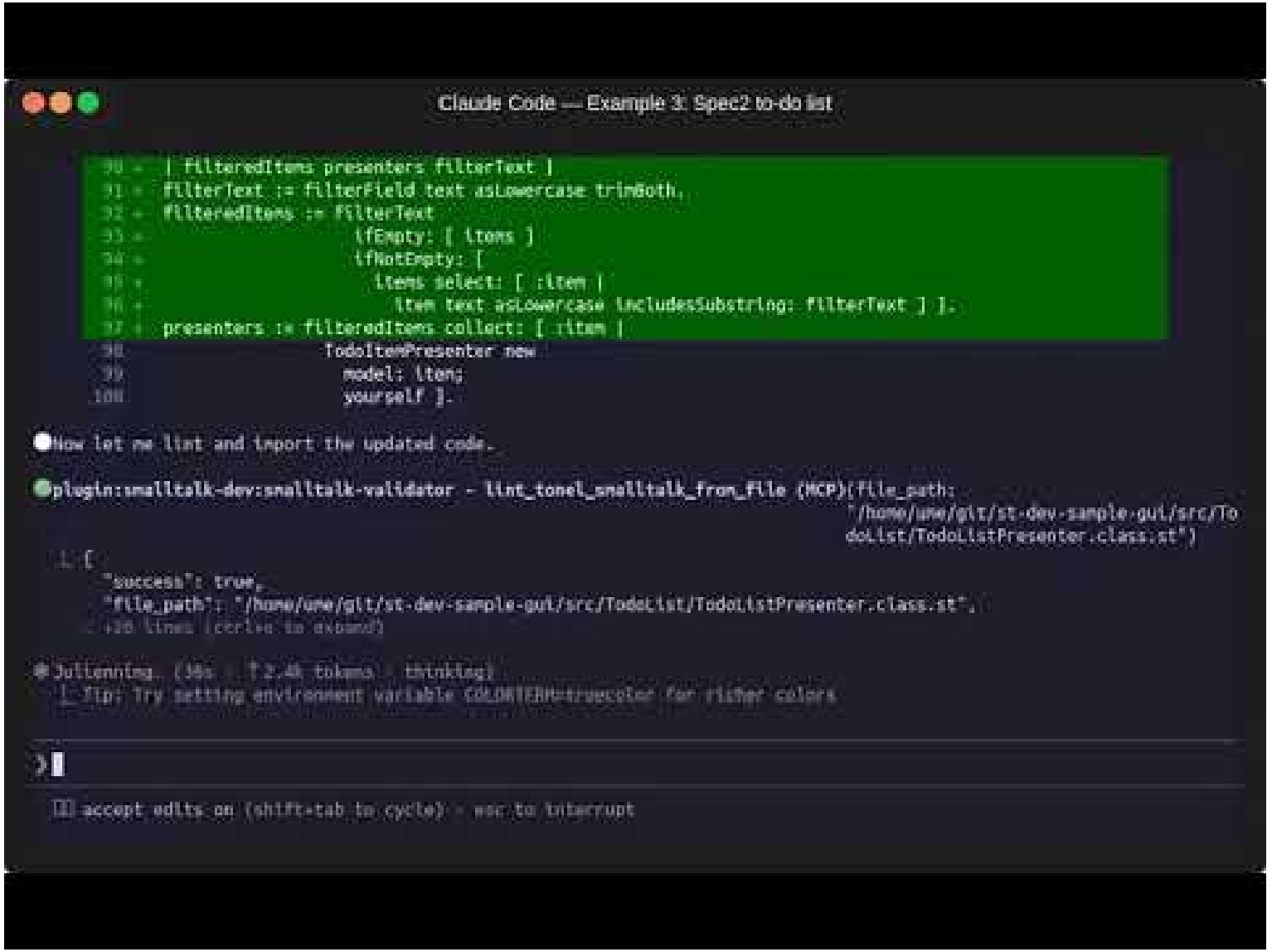
- Video
- Generated source
- Demo

Example 3 — GUI Programming

Building a GUI application with Spec2.

```
/st-buddy I want to make a simple to-do list using the Spec2 framework.
Include a checkbox for each item, plus an input field
with Add/Remove buttons at the bottom.
Only checked items can be removed. Start developing.
```

Example 3 — What Happens Next



- Video
- Generated source
- Demo

Case Studies

Case Studies — Real Projects

Projects created using smalltalk-dev-plugin.

> Both were **over 90% built with Claude Code + smalltalk-dev-plugin**.

pharo-acp

A library for connecting to AI agents from the Pharo side using the ACP protocol.

Connects to: Gemini CLI, Claude Code, OpenCode, Copilot CLI, Codex CLI, etc.

https://github.com/mumez/pharo-acp

pharo-acp-chat-ui

A simple AI Chat GUI for Pharo. Uses pharo-acp internally.

https://github.com/mumez/pharo-acp-chat-ui

Summary

smalltalk-dev-plugin makes Pharo Smalltalk development with AI assistants practical and productive.

Just type `/st-buddy` and start building.

Feedback and contributions are welcome!

https://github.com/mumez/smalltalk-dev-plugin